

Final Consulting Project Report

Cheese Aroma liking

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1 Summary

This project focused on identifying chemical compounds associated with plant-based cheese aroma and flavor liking using high-dimensional chemical composition data and consumer sensory ratings, while also examining consumer-level factors related to plant-based cheese preference. The dataset included 20 cheese samples with 4 replicates each and 135 consumers, with each consumer evaluating every sample. In addition to modeling chemical predictors of liking, I analyzed open-ended consumer responses using text analysis and word clouds to better understand the attributes associated with positive and negative reactions to specific cheese samples.

To study differences in consumer behavior, I used demographic and dietary information and measures of consumer inclination toward dairy cheese to explore patterns in preference through clustering and principal component analysis. For the chemical analysis, I applied a range of high-dimensional statistical and machine learning methods, including Elastic Net, Sure Independence Screening with Pearson and distance correlation, sparse partial least squares, and XGBoost. The final chemical rankings were obtained using an ensemble approach that aggregated results across methods to identify compounds most consistently associated with aroma and flavor liking.

2 Client Background and Project Objectives

2.1 Client Background

In previous analyses, the client used OPLS, Pareto transformations, and statistical methods such as ANOVA, mixed-effects models, and correlation analysis to study both the chemical and consumer-level sensory data. My role in this project was to support the exploration of additional modeling approaches, analyze consumer-level data to identify factors associated with liking and disliking plant-based cheeses, and provide statistical feedback on the validity and interpretation of the client's existing analyses.

2.2 Problem Statement

The client sought to identify chemical compounds associated with plant-based cheese flavor and aroma liking in order to better understand which compounds may contribute to desirable sensory profiles. In addition to the chemical analysis, the project also aimed to identify consumer-level characteristics associated with liking or disliking particular plant-based cheese samples and flavor profiles.

The data included high-dimensional chemical composition measurements for 20 cheese samples, along with consumer sensory evaluations collected from 135 participants. Each consumer evaluated all cheese samples, which resulted in a rich set of repeated sensory responses as well as consumer-level information that could be used to study variation in preferences across individuals.

The primary analytical challenge was twofold. First, the chemical data involved a large number of predictors relative to the number of cheese sample, requiring the use of high-dimensional statistical and machine learning methods to identify compounds associated with aroma and flavor liking. Second, the consumer-level data exhibited substantial heterogeneity in preferences, making it important to examine whether demographic factors, prior inclination toward dairy cheese, dietary and lifestyle information, and qualitative consumer responses could help explain differences in liking across samples.

Accordingly, the project focused on combining chemical-level modeling with consumer-level analysis. The goal was not only to identify compounds associated with plant-based cheese liking, but also to better understand which consumer characteristics and response patterns were linked to positive or negative perceptions of specific cheese products.

2.3 Project Objectives

The objectives of this project were to:

- Identify chemical compounds associated with plant-based cheese aroma and flavor liking using the available omics and sensory data.
- Evaluate high-dimensional statistical and machine learning methods for screening and ranking predictive chemical compounds.
- Combine results across multiple modeling approaches to produce an aggregate ranking of chemicals most consistently associated with aroma and flavor liking.
- Analyze consumer-level data to identify demographic, behavioral, and preference-related factors associated with liking or disliking plant-based cheese samples.
- Explore qualitative consumer responses using text analysis and word clouds to better understand attributes linked to positive and negative perceptions of specific cheeses.
- Investigate heterogeneity in consumer preferences through clustering and principal component analysis to identify groups of consumers with similar patterns of liking and cheese preference.

3 Deliverables

Deliverables were provided to the client iteratively throughout the semester. After each method was implemented, I met with the client to share the code, discuss the results, explain their interpretation, and review the strengths and limitations of the approach.

These ongoing deliverables included analysis code, method-specific results, visualizations, and explanation of the methods employed. I also showed the client how to reproduce the analyses and explained the motivation behind the suggested methods.

4 Data

The project used OMICs data together with consumer-level sensory and demographic information. The OMICs dataset contained measurements on 50 chemical predictors for the plant-based cheese samples. These chemical measurements were used to study associations between chemical composition and consumer evaluations of aroma and flavor liking.

To support different parts of the analysis, the data were organized into three related datasets. The first was a chemical-level dataset containing the 50 chemical predictors for the cheese samples. The second was a consumer-level dataset containing sensory responses and consumer-specific information, including variables used to study differences in preference across individuals. The third was a combined dataset created by joining the chemical and consumer-level data, allowing the analysis to relate sample composition to consumer responses at the individual level.

These datasets were used for complementary purposes throughout the project. The chemical-level data supported high-dimensional modeling aimed at identifying compounds associated with aroma and flavor liking, while the consumer-level and combined datasets were used to examine patterns in consumer preferences, demographic variation, and other factors associated with liking or disliking plant-based cheese samples.

5 Methods

The analysis combined text analysis, multivariate methods, mixed-effects modeling, and high-dimensional supervised learning to study plant-based cheese aroma liking. Consumer comments were first analyzed using standard text preprocessing, TF-IDF scoring, and word clouds to summarize positive and negative feedback for individual cheese samples. To examine structure in consumer preferences, tetrachoric correlations, hierarchical clustering, and factor analysis were used to study patterns in dairy cheese liking and to derive lower-dimensional latent preference factors. Exploratory analysis of the chemical data was then conducted using log transformation, principal component analysis, and hierarchical clustering to assess broad correlation structure and identify groups of chemicals with similar behavior across samples.

To identify chemicals associated with aroma liking, I first fit a mixed-effects model to adjust the consumer-level aroma ratings for subject-specific variation and measured consumer characteristics. The residualized aroma responses were then linked to the chemical predictors and analyzed using several high-dimensional screening methods, including Elastic Net regression, Sure Independence Screening based on Pearson and distance correlation, sparse partial least squares, and XGBoost. To combine information across methods, I used a bootstrap-based stability procedure in which each method was repeatedly refit across resampled datasets, and method-specific rankings were weighted according to their stability. A final composite ranking was then constructed to identify the chemicals most consistently associated with aroma liking across the screening approaches.

6 Findings

The bootstrap-based ensemble screening identified several chemical variables as the most stable predictors of adjusted aroma liking. The highest-ranked chemicals according to the composite score were Feature30_N, Feature9_N, Feature46_N, Feature1_N, Feature42_N, Feature23_N, Feature6_N, Feature36_N, Feature45_N, and Feature34_N. These variables were selected because they received relatively strong support across multiple screening methods and remained competitive under repeated bootstrap resampling.

While the individual methods did not produce identical rankings, the composite procedure helped identify chemicals that were consistently important rather than highly ranked by only a single model. This was especially valuable in the present high-dimensional setting, where different methods capture different forms of association and may respond differently to correlated predictors. Overall, the final ranking provided a more robust summary of which chemicals were most plausibly associated with aroma liking after adjusting for consumer-level effects.

7 Future Improvements

Several directions could improve and extend the present analysis. First, additional refinement of the XGBoost model may be worthwhile, since its rankings showed the greatest disagreement with the other screening methods. In particular, future work could explore alternative pruning strategies, stronger regularization, and more systematic tuning of tree complexity in order to reduce overfitting and improve the stability of the resulting feature rankings.

Second, the current analysis treated each chemical variable individually, but future work could consider incorporating differential penalties or other structured regularization schemes that reflect prior information about the likely predictive importance of specific classes of chemicals. This may improve both interpretability and stability, especially in the presence of strong correlation among predictors.

Another important direction is to incorporate consumer-specific heterogeneity more directly into the modeling framework. Although the current analysis adjusted for subject-level effects through residualization, future work could explore models that allow chemical effects to vary across consumer groups or individual preference profiles. This may help capture idiosyncratic differences in sensory response and provide a more detailed understanding of how different types of consumers perceive plant-based cheese aroma and flavor.

Finally, it may be useful to move beyond the raw chemical variables and instead construct higher-level aroma or flavor profiles using dimension reduction and clustering methods informed by subject-matter knowledge. Alternative clustering approaches, as well as modified PCA-based or factor-analytic representations of the chemical data, may yield more interpretable latent flavor profiles that better align with sensory science considerations and improve downstream prediction.

A Technical Appendix

A.1 Detailed Methodology

A.1.1 Data Cleaning and Transformations

The analysis used three related datasets constructed from the original study files: a subject-level dataset, a cheese-level chemical dataset, and a rating dataset containing repeated cheese–consumer interactions. The subject-level dataset contained one row per consumer and included demographic variables, dietary variables, prior exposure to plant-based cheese, and indicators of dairy cheese preference. The cheese-level dataset contained one row per cheese sample replicate with the measured OMICS variables, and the rating dataset contained consumer-specific aroma ratings and related sensory responses for each cheese evaluation.

Missing values in the subject-level data were handled using within-subject imputation when the missing variables represented stable subject characteristics. In particular, missing cheese preference values were imputed using the corresponding preference responses from the same subject recorded for other cheese evaluations, since these variables were intended to describe persistent consumer characteristics rather than cheese-specific responses. Missing values in the chemical data for Feature47 were imputed using the average observed value of that feature across the available cheese samples. After cleaning, the three datasets were retained separately for downstream analysis and merged only when needed for specific modeling tasks.

For the supervised chemical screening analysis, chemical predictors were log transformed to reduce the effect of strong dispersion and to stabilize the scale of the predictors. The transformed chemical variables were then standardized before fitting high-dimensional screening methods. On the response side, aroma liking was treated as a continuous outcome measured on a 1–9 hedonic scale. This approximation was adopted to simplify modeling and screening, with the expectation that it would not materially alter the main inferential conclusions in the present setting.

A.1.2 Word Cloud Construction

Open-ended consumer feedback was analyzed separately for positive and negative comments for each cheese sample. The raw text responses were first cleaned using standard preprocessing steps, including removal of punctuation, stop words, and other non-informative tokens. Spelling was corrected where necessary, and the text was standardized so that common words and phrases would be represented consistently across responses.

After preprocessing, the text data were vectorized and term frequency-inverse document frequency (TF–IDF) scores were computed for the terms in the corpus. TF–IDF was used to identify words that were not only frequent, but also relatively informative within a given collection of comments. Word clouds were then created as a descriptive visualization tool, with larger words corresponding to more prominent or informative terms. These visualizations were used to summarize which

attributes were most commonly associated with favorable and unfavorable responses to particular cheese samples.

A.1.3 Consumer Data Analysis

The subject-level data were used to study whether pre-existing dairy cheese preferences could help explain variation in consumer responses to plant-based cheese. Consumers indicated whether they liked each of 10 dairy cheese types, yielding a set of binary preference variables. Because these variables were sparse and highly structured, two approaches were considered for dimension reduction and interpretation: hierarchical clustering based on the tetrachoric correlation matrix and factor analysis applied to the same matrix.

Tetrachoric correlations were used because the cheese preference variables were dichotomous and were viewed as coarse indicators of underlying latent preferences. Hierarchical clustering was then applied to the tetrachoric correlation matrix using a distance of $1-r$, where r denotes the tetrachoric correlation. This was intended to determine whether certain dairy cheeses tended to be liked together and could therefore be grouped into interpretable preference profiles. Although clustering provided some exploratory insight, the resulting groups were not especially strong or stable.

Because clustering alone did not yield sufficiently clear composite structures, factor analysis was also applied to the tetrachoric correlation matrix. Based on parallel analysis, a two-factor solution was selected and used to construct latent scores summarizing subject-level cheese preferences. These latent variables, denoted Cheese_MR1 and Cheese_MR2, replaced the original set of cheese preference indicators in later modeling. In addition, frequency of plant-based cheese consumption was retained as the main subject-level behavioral covariate, while other related frequency and exposure variables were excluded to reduce redundancy. The final subject-level covariates used in the downstream residualization step were Unique Panelist ID, Frequency_of_PB_Cheese, Age, Cheese_MR1, and Cheese_MR2.

A.1.4 Chemical Data Structure

Before attempting to identify predictive chemicals, the chemical data were analyzed to determine whether there was meaningful lower-dimensional structure among the OMICS variables. Because the chemical measurements showed substantial dispersion, all chemical predictors were log transformed prior to exploratory analysis. Principal component analysis (PCA) was then used to summarize the dominant sources of variation in the transformed chemical design matrix.

Based on the scree plot, the first 8 principal components were retained for interpretation, and these components explained more than 90% of the total variation in the transformed chemical data. Scatterplots of the leading principal components were used to visually assess whether similar cheese samples tended to cluster together in the reduced-dimensional space. Biplots and loading summaries were also examined in order to identify which chemicals contributed most strongly to the principal directions of variation. Although some local structure was observed, the PCA results suggested

substantial shared variance and cross-loading among chemicals rather than sharply separated latent groups.

To complement the PCA, hierarchical clustering was also applied to the chemical data after transposing the transformed matrix so that each chemical, rather than each cheese sample, was treated as an observational unit. This was done to examine whether chemicals with similar profiles across cheese samples could be grouped together. Several candidate clustering solutions were explored, and although none produced especially strong separation, a four-cluster solution was retained as a reasonable descriptive summary. The PCA and clustering analyses were not used directly as predictive tools, but rather as exploratory methods to understand correlation structure and to aid interpretation of the later screening results.

A.1.5 Residualization and Ensemble Screening

The primary supervised objective of the analysis was to identify cheese-level chemicals associated with aroma liking after adjusting for subject-level heterogeneity and measured consumer characteristics. To accomplish this, a mixed-effects model was first fit to the consumer-level aroma ratings with a random intercept for subject and fixed effects for Frequency_of_PB_Cheese, Age, Diet, Cheese_MR1, and Cheese_MR2:

$$Y_{ij} = \beta_0 + \beta_1 \text{FreqPB}_i + \beta_2 \text{Age}_i + \beta_3 \text{Diet}_i + \beta_4 \text{CheeseMR1}_i + \beta_5 \text{CheeseMR2}_i + b_i + \varepsilon_{ij},$$

$$b_i \sim N(0, \sigma_b^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2),$$

This step was intended to remove systematic variation in aroma ratings attributable to measured consumer-specific factors and persistent between-subject differences.

The residuals from this mixed-effects model were then treated as adjusted aroma liking responses:

$$e_{ij} = Y_{ij} - \hat{Y}_{ij},$$

where

$$\hat{Y}_{ij} = \hat{\beta}_0 + \hat{\beta}_1 \text{FreqPB}_i + \hat{\beta}_2 \text{Age}_i + \hat{\beta}_3 \text{Diet}_i + \hat{\beta}_4 \text{CheeseMR1}_i + \hat{\beta}_5 \text{CheeseMR2}_i + \hat{b}_i.$$

These residualized responses were averaged at the replicate level and merged with the corresponding chemical measurements so that the remaining analysis could focus on identifying chemicals associated with unexplained variation in aroma liking. The chemical predictors in this final screening dataset were log transformed and standardized before fitting the high-dimensional methods.

Several screening methods were then used to rank chemicals according to their association with

the adjusted aroma response. These included Elastic Net regression, Sure Independence Screening (SIS) using Pearson correlation, SIS using distance correlation, sparse partial least squares, and XGBoost. Elastic Net was used with a fixed specification of $\alpha = 0.5$ and $\lambda = \exp(-2.8)$. Sparse partial least squares was fit using $ncomp = 2$ and $keepX = (15, 20)$. XGBoost was fit using a fixed set of parameters selected during earlier model development. Each method produced a ranking of the chemical predictors based on its corresponding importance measure.

Because the different methods do not necessarily agree, and because the data are high-dimensional relative to the number of cheese observations, a bootstrap-based stability procedure was used to quantify the robustness of the rankings. Across repeated bootstrap samples, each screening method was refit and the rank of every chemical was recorded. For each method, the standard deviation of the rank of each chemical across bootstrap samples was computed, and these standard deviations were summarized to obtain a method-level instability measure. Methods that produced more stable rankings were given larger weights, while methods with greater variability were downweighted.

The final ensemble ranking was constructed by combining the average ranks across methods using weights derived from their bootstrap stability. In this way, the final composite score emphasized chemicals that were not only ranked highly, but also ranked consistently across repeated samples and across multiple screening approaches. This ensemble procedure was intended to reduce reliance on any single model and to provide a more robust summary of which chemicals were most plausibly associated with aroma liking.

A.2 Methodological Results and Figures

A.2.1 Word Cloud Findings

The word cloud analysis provided a descriptive summary of the language consumers used when discussing the most- and least-liked cheese samples. For highly rated samples such as C and K, the positive reviews contained terms suggesting more favorable sensory associations, including words such as “balanced,” “nutty,” “roasted,” “rich,” and “nice.” As shown in Figure 1, these samples generally elicited descriptors that were more consistent with pleasant, savory, or cheese-like sensory characteristics.



Figure 1: Word clouds for positive consumer reviews of Samples C and K, two of the more highly liked cheeses. The dominant terms suggest comparatively favorable aroma and flavor associations.

In contrast, the least-liked samples, I and J, were associated with strongly negative descriptors in their negative review word clouds. As shown in Figure 2, terms such as “spoiled,” “putrid,” “garbage,” “rotten,” and “unappetizing” appeared prominently, suggesting that these samples were often perceived as sour, stale, fermented, chemically unpleasant, or otherwise off-putting. These descriptors indicate that disliked samples were not merely viewed as weak or bland, but were often associated with strong sensory defects or aversive aroma characteristics.

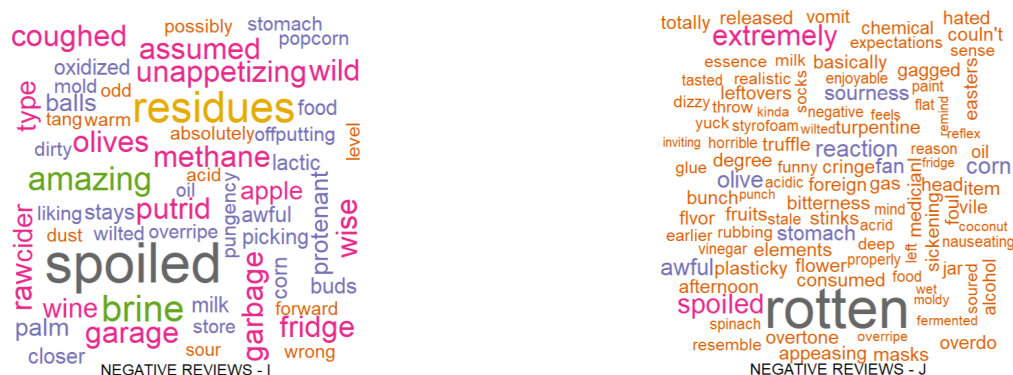


Figure 2: Word clouds for negative consumer reviews of Samples I and J, the least-liked cheeses. The dominant language reflects strong negative sensory reactions and perceptions of spoilage or chemical off-notes.

An important qualitative finding was that even favorable comments for poorly rated samples tended to be weaker and less clearly positive than the comments observed for better-performing cheeses. Overall, the text analysis supported the quantitative liking results by showing that well-liked cheeses such as C and K were associated with more balanced and savory descriptors, whereas poorly liked cheeses such as I and J were consistently linked to spoilage-related, chemical, and unpleasant sensory language.

A.2.2 Consumer Inclination to Dairy Cheese

To study whether pre-existing dairy cheese preferences could help explain variation in consumer responses to plant-based cheese, binary indicators of whether a subject liked each of 10 dairy cheese types were analyzed. Because these variables were dichotomous and intended to reflect underlying latent preference tendencies, tetrachoric correlations were computed to assess the dependence structure among the cheese preference variables. The resulting correlation matrix, shown in Figure 3, suggested broadly positive association across many cheese types, with some evidence of stronger local groupings rather than sharply separated preference profiles.

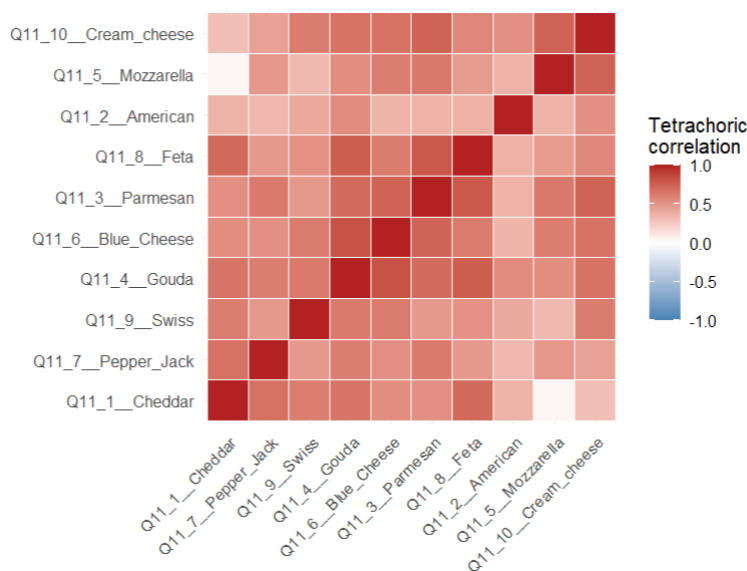


Figure 3: Heatmap of the tetrachoric correlation matrix for dairy cheese preference indicators. Most cheese types show positive association, although the overall structure is diffuse and does not suggest sharply separated groups.

To further investigate possible grouping structure, hierarchical clustering was applied to the tetrachoric correlation matrix using a distance of $1 - r$, where r denotes the tetrachoric correlation. The resulting dendrogram is shown in Figure 4, and a four-cluster solution was explored for interpretability. This solution suggested that cheddar and pepper jack formed one cluster, American cheese formed its own cluster, Parmesan, Gouda, Mozzarella, Blue Cheese, Feta, and Cream Cheese formed a larger third cluster, and Swiss formed a fourth cluster by itself. While these groupings provided some exploratory insight, the clustering structure was not especially strong, and the resulting clusters were not sufficiently distinct for direct use as interpretable subject-level covariates.

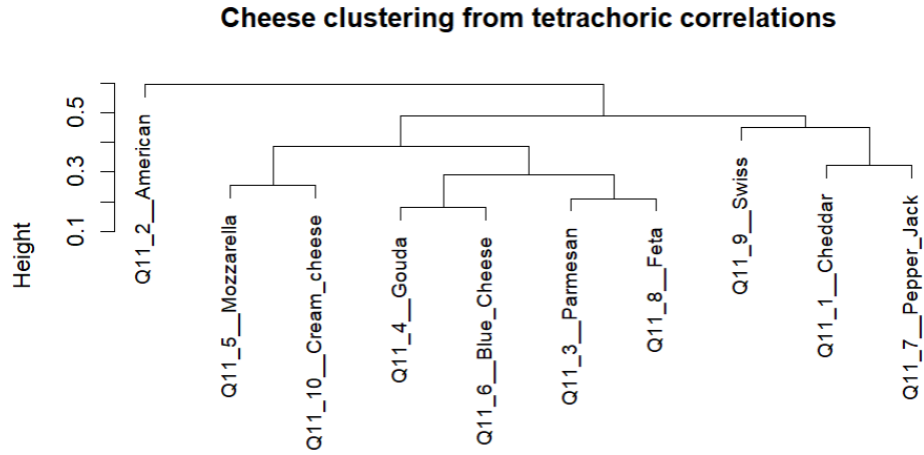


Figure 4: Hierarchical clustering dendrogram based on tetrachoric correlations among dairy cheese preference indicators. The clustering suggests some local similarity among cheeses, but the overall grouping structure is not especially strong.

Because the clustering analysis did not yield sufficiently clear or stable preference groups, factor analysis was applied to the tetrachoric correlation matrix in order to construct lower-dimensional latent variables summarizing dairy cheese inclination. Parallel analysis, shown in Figure 5, supported the use of a two-factor solution. A three-factor solution was also considered, but substantial correlation among the resulting factors and the tendency of cheddar to dominate one factor made the higher-dimensional representation less attractive for interpretation.

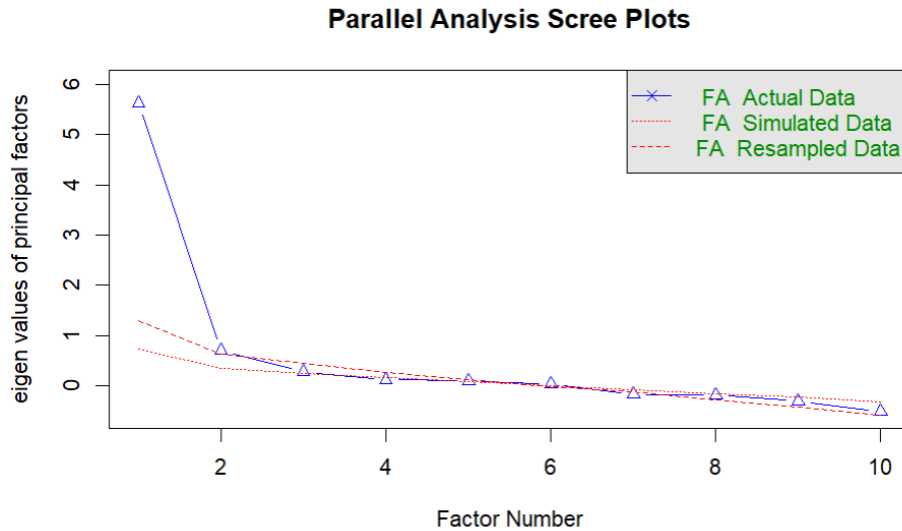


Figure 5: Parallel analysis scree plot used to determine the number of latent factors for dairy cheese preference variables. The results supported retaining two latent factors.

The resulting factor structure was easier to use for downstream modeling than the raw set of cheese preference indicators, although it was still not especially clean from an interpretive standpoint. Mozzarella and Cream Cheese loaded very strongly on the first latent factor, while Parmesan, Gouda, and Blue Cheese also showed moderately strong loadings on this factor. Cheddar loaded very strongly on the second latent factor, while Gouda, Pepper Jack, Feta, and Swiss showed moderate cross-loadings on both factors. Because of these cross-loadings, the factors did not correspond neatly to sharply distinct cheese categories.

Substantively, the first latent factor appeared to reflect a broader openness to creamy and somewhat distinctive cheeses, while the second latent factor appeared to correspond more closely to firmer, savory, and more commonly consumed cheeses. However, this interpretation should be treated cautiously. In particular, Gouda, Swiss, Pepper Jack, and Feta were mixed items rather than pure indicators of a single latent dimension, and the second factor was heavily influenced by cheddar, which was liked by a large majority of subjects. For this reason, the latent factors were primarily used as dimension-reduced subject-level covariates rather than as sharply interpretable sensory constructs.

Overall, although the consumer dairy cheese preference variables did not exhibit especially strong or clean structure, the factor-analytic summary provided a practical way to incorporate subject-level cheese inclination into the later residualization and predictive screening analyses.

A.2.3 Exploratory Analysis of Chemical Structure

Before conducting supervised screening, the chemical predictors were analyzed to determine whether there was meaningful lower-dimensional structure in the OMICS data. Because the chemical variables showed substantial dispersion, all predictors were first log transformed. Principal component analysis (PCA) was then used as an exploratory tool to summarize the dominant directions of variation in the transformed design matrix and to assess whether cheese samples with similar chemical compositions tended to cluster together.

The first two principal components are shown in Figure 6, where each point corresponds to a sample–replicate combination. The sample labels make it possible to assess whether replicates of the same cheese tend to appear near one another in the reduced-dimensional space. Several cheese samples appeared to form local groupings, suggesting that at least some of the chemical variation was structured by sample identity rather than being entirely random. At the same time, the separation between groups was not uniformly sharp, indicating that the chemical composition of the cheeses exhibited both within-sample similarity and substantial shared variation across products.

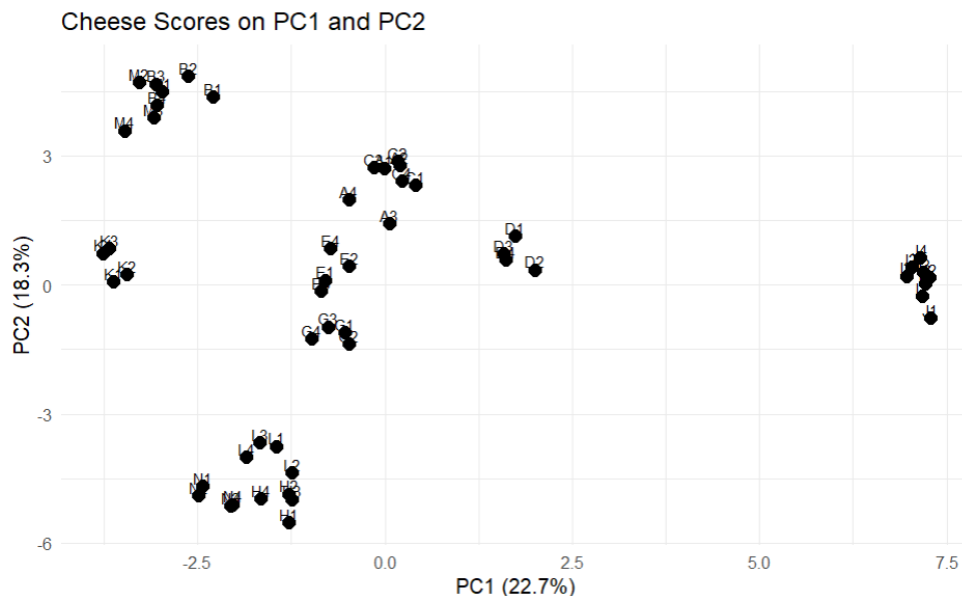


Figure 6: Scores plot of cheese sample replicates on the first two principal components. Sample labels identify replicate-level observations and allow visual assessment of which cheeses have similar chemical profiles.

To better understand which chemicals contributed to the major directions of variation, the PCA biplot was examined, as shown in Figure 7. This visualization suggested that many chemicals contributed jointly to the first two principal components, with substantial overlap in loading directions rather than a small number of isolated dominant features. In other words, the leading components appeared to reflect combinations of correlated chemicals rather than clean, one-to-one mappings between individual compounds and principal axes.

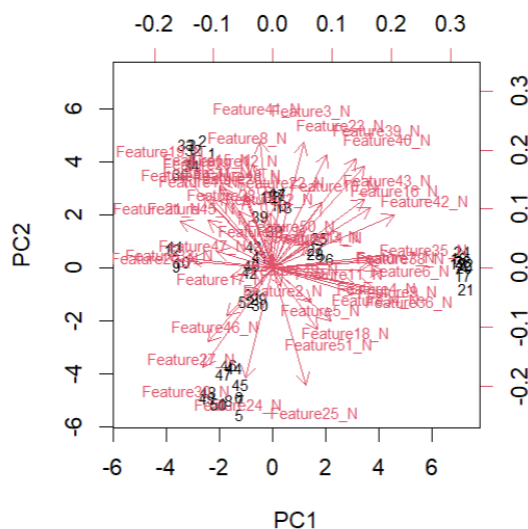


Figure 7: PCA biplot for the chemical predictors. The arrows represent chemical loadings on the first two principal components and illustrate that many chemicals contribute jointly to the dominant directions of variation.

To complement the PCA, hierarchical clustering was also applied to the log-transformed chemical matrix after transposing the data so that chemicals, rather than cheese samples, were treated as the observational units. This was done to examine whether chemicals with similar behavior across the cheese samples could be grouped together. Although several clustering solutions were explored, the resulting structure was not especially strong, and the clustering was treated primarily as an exploratory aid rather than a definitive partition of the chemical variables.

The relationship between the exploratory chemical clusters and the principal components is summarized in Figure 8. This figure shows how chemicals within the identified clusters loaded across the leading principal components. While some clusters appeared to show similar loading patterns, there was also substantial heterogeneity within clusters and evidence of shared variance across groups. This reinforced the impression that the chemical variables were not organized into a small number of sharply separated blocks, but instead exhibited overlapping correlation structure.

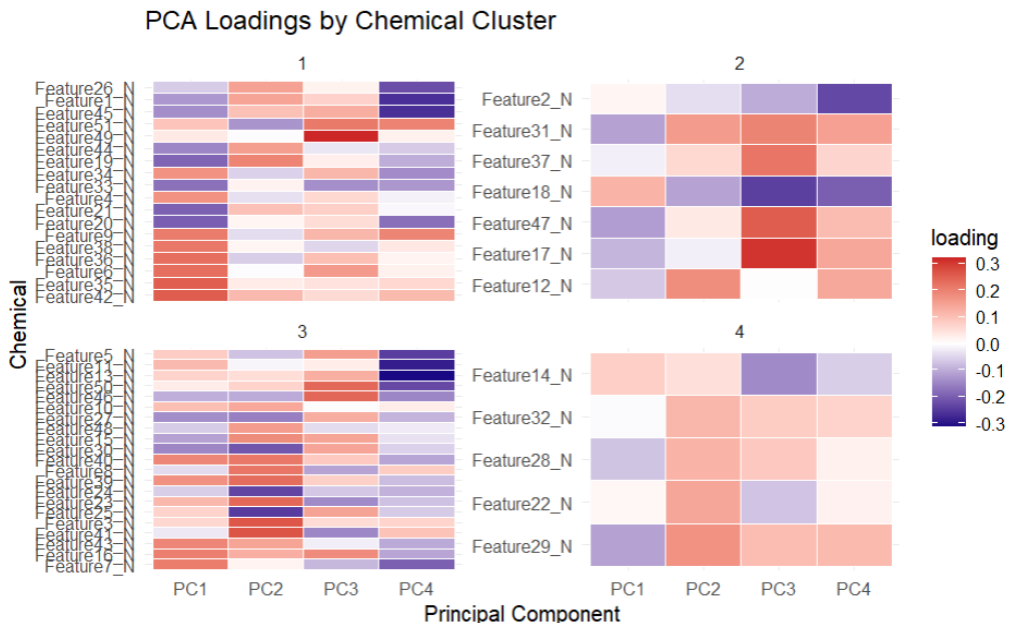


Figure 8: Heatmap of PCA loadings grouped by exploratory chemical cluster. The results suggest some similarity in loading patterns within clusters, but also substantial overlap and cross-structure across components.

Diagnostic summaries such as the clustering dendrogram and silhouette widths were also examined, but these supported the conclusion that the clustering structure was only moderately informative and should be interpreted cautiously.

Overall, the exploratory analysis suggested that the chemical predictors contained meaningful structure, but not in a way that yielded cleanly separated or easily interpretable groups. PCA indicated that much of the variation could be summarized in a lower-dimensional space and that some cheese samples had similar chemical profiles, while hierarchical clustering suggested only moderate grouping among the chemicals themselves. These findings motivated the use of multiple high-dimensional screening methods in the later analysis, since no single low-dimensional or cluster-based representation appeared sufficient to fully capture the relationship between chemical composition and aroma liking.

A.2.4 Supervised Learning Ensemble Approach

Following the exploratory analysis of the chemical predictors, a supervised ensemble screening strategy was used to identify the compounds most strongly associated with aroma liking. Because the predictor space was high-dimensional and different learning procedures can emphasize different aspects of the relationship between predictors and response, no single screening method was treated as definitive. Instead, several complementary procedures were applied, namely sure independence screening (SIS), distance-correlation sure independence screening (DC-SIS), elastic net, sparse partial least squares (sPLS), and XGBoost. Taken together, these methods provided

a range of linear, nonlinear, regularized, and dimension-reduction-based approaches for identifying candidate chemical predictors.

As an initial diagnostic step, the top-ranked chemicals were compared across methods using their average ranks. The resulting heatmap, shown in Figure 9, was used primarily as a sanity check and as a way to assess broad agreement among the candidate screening procedures before constructing the final ensemble ranking. Several chemicals appeared near the top across multiple methods, which suggested that at least part of the signal was stable across modeling strategies. At the same time, noticeable disagreement was present for some variables, especially for XGBoost and, to a lesser extent, elastic net, indicating that different methods were capturing somewhat different aspects of the predictor-response relationship.

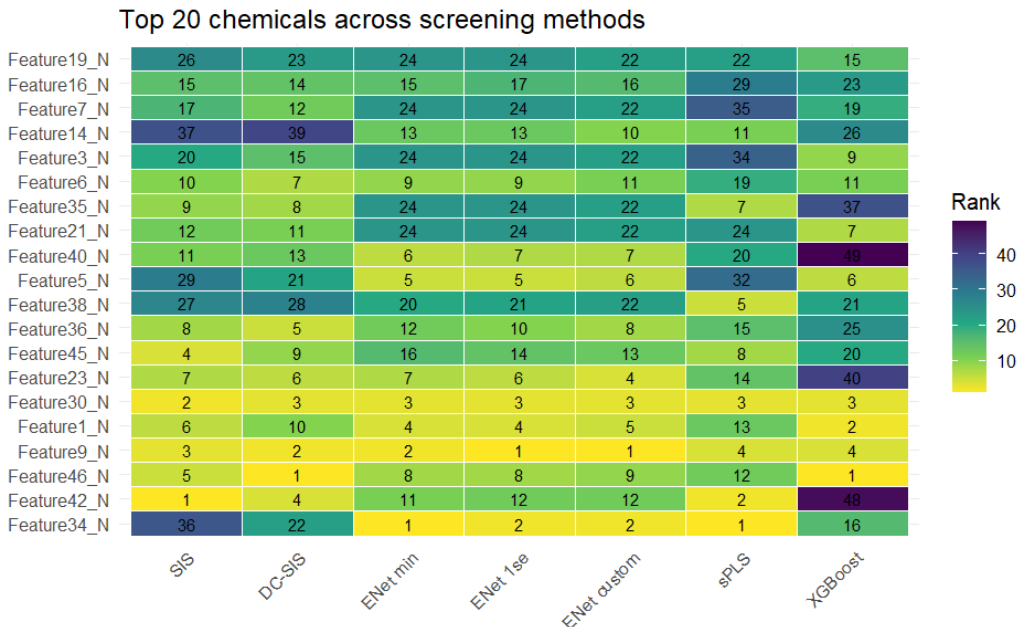


Figure 9: Heatmap of average ranks for the top 20 candidate chemicals across screening methods. This plot was used as an exploratory diagnostic and sanity check to assess broad agreement across methods before forming the final ensemble ranking.

Because variable rankings obtained from a single fitted model can be sensitive to sample variation, a bootstrap-based resampling strategy was used to stabilize the screening results. Across repeated bootstrap resamples of the subjects, each screening method was refit and the resulting variable rankings were recorded. This produced an empirical distribution of ranks for each chemical under each method, allowing both the central tendency and the stability of the rankings to be assessed. The bootstrap procedure was especially useful in this setting because it reduced the risk of over-interpreting rankings that might arise from idiosyncrasies of a single sample.

For elastic net, hyperparameter tuning was not based solely on mechanical minimization of cross-validated prediction error. Instead, the cross-validation error curve was used as a diagnostic guide, and the final choice of λ was informed by visual inspection of a stable low-error region in the

solution path, as shown in Figure 10. This choice was made to avoid over-emphasizing negligible differences in cross-validation error and to favor a regularization level that produced a more stable and interpretable screening result. Thus, cross-validation was used to inform the tuning decision, but not to determine it automatically in a purely algorithmic way.

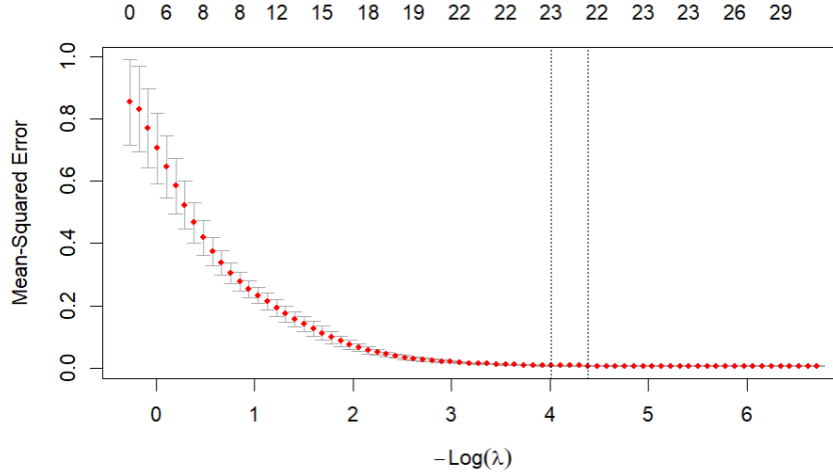


Figure 10: Cross-validated error curve for elastic net tuning. Rather than relying exclusively on the numerical minimizer, the final regularization level was chosen by visually identifying a stable low-error region of the curve.

To combine the results of the screening procedures into a single ensemble ranking, method-specific weights were constructed to reflect ranking stability under bootstrap resampling. Rather than assigning equal importance to all methods, the weighting scheme rewarded methods that ranked variables more consistently across repeated bootstrap samples and down-weighted methods with greater uncertainty. The weighting procedure was defined as follows:

- Each screening method was run over 30 bootstrap resamples of the data.
- For each method, the variability of its ranking behavior across bootstrap runs was summarized using the standard deviation of the ranks.
- This standard deviation served as a measure of instability: methods with smaller standard deviations were regarded as more stable, while methods with larger standard deviations were regarded as less reliable.
- A method-specific weight was then assigned according to

$$w_j = \exp(-ks_j),$$

where s_j denotes the bootstrap-based standard deviation for method j , and $k = 0.08$.

- These weights were then normalized and used to compute a weighted average of the method-specific ranks for each chemical.

Under this rule, methods with more stable ranking behavior retained greater influence in the final aggregation, while unstable methods were down-weighted exponentially. The resulting composite score therefore reflected not only how highly a chemical was ranked on average, but also how reliably those rankings were reproduced across resamples.

The method-specific bootstrap rankings for the final shortlisted chemicals are shown in Figure 11. Each line corresponds to one screening method and displays how that method ranked the selected chemicals. This plot makes it possible to distinguish chemicals that were consistently prioritized across methods from those whose apparent importance depended more strongly on a particular modeling framework. In particular, SIS, DC-SIS, and sPLS showed relatively stronger agreement with one another, whereas XGBoost displayed more pronounced deviations and elastic net also differed moderately for several chemicals. This pattern suggests that some chemicals were robust signals across a variety of screening assumptions, while others were more sensitive to nonlinear effects or penalized regression structure.

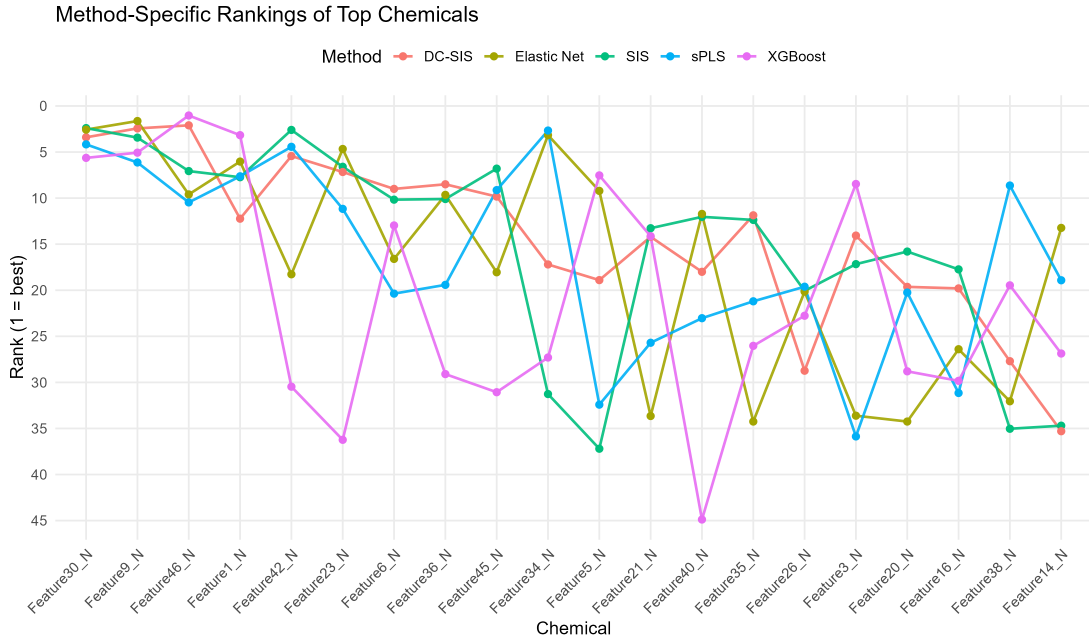


Figure 11: Bootstrap-based method-specific rankings of the final 20 shortlisted chemicals. Agreement among SIS, DC-SIS, and sPLS is visible, while XGBoost and, to a lesser extent, elastic net show greater disagreement for several chemicals.

The final ensemble ranking obtained from the weighted average of the bootstrap-based ranks is displayed in Figure 12. This plot gives the overall ordering of the selected chemicals after combining evidence across all five screening procedures. Chemicals with lower composite ranks were interpreted as receiving stronger and more stable overall support from the ensemble. Because the

final ranking incorporated both average importance and method stability, it provided a more robust basis for prioritization than any individual screening method alone.

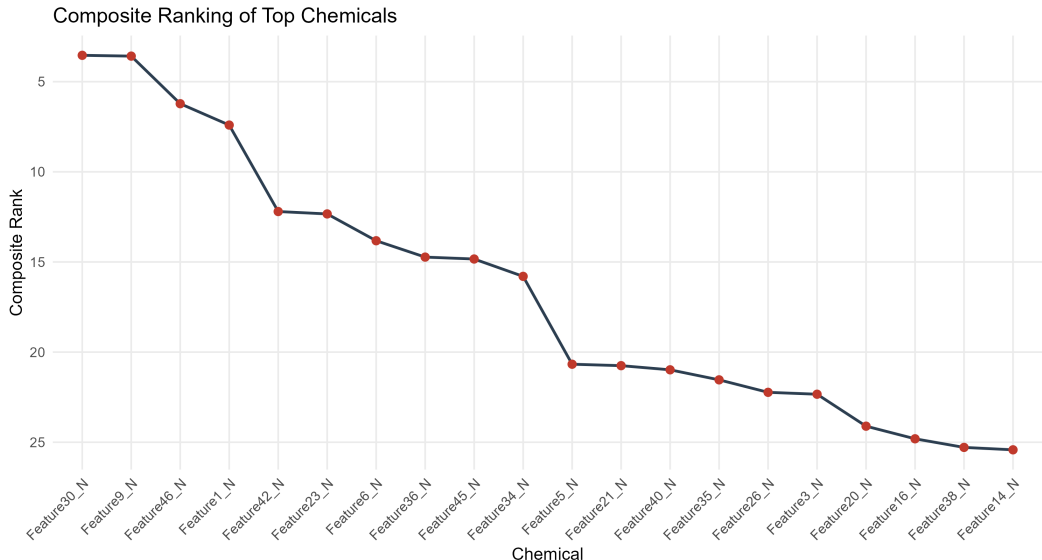


Figure 12: Final composite ranking of the shortlisted chemicals obtained by aggregating bootstrap-based method-specific ranks using stability-based weights. Lower composite rank indicates stronger overall support from the ensemble.

Overall, the supervised ensemble screening approach identified a set of chemicals that repeatedly appeared near the top of the rankings across multiple methods and bootstrap resamples. The ensemble strategy was useful because it reduced dependence on any single screening procedure and provided some protection against method-specific instability. In particular, the visible disagreement of XGBoost and, to a lesser extent, elastic net with the other methods reinforced the value of combining multiple screening perspectives rather than relying on one algorithm in isolation. The final composite ranking was therefore used as the principal basis for selecting candidate chemicals for the downstream analysis.